

Technology, Tasks, And Labor-Market Adjustment

Core Frameworks

Special Topic 5: Technology, Innovation, and Labor – Week 1

Central question

- How should labor economists map technologies into tasks, skills, labor demand, worker adjustment, and equilibrium outcomes?
- Technology is not the outcome. Wages, employment, mobility, organization, inequality, and welfare are the labor outcomes.
- Week 1 discipline: define the technology object before interpreting a technology coefficient.

Why this is labor economics

- Technology changes what firms want workers to do.
- Innovation changes which skills are valuable and which tasks can be performed by labor, capital, software, or teams.
- Adoption changes hiring, training, supervision, workflow, and job design inside firms.
- Equilibrium adjustment determines who gains, who loses, and which effects persist.

Four margins to separate

Margin	Labor-market meaning
Invention	a new capability appears
Adoption	a firm, establishment, or worker begins using it
Diffusion	adoption spreads across firms, places, occupations, or workers
Reorganization	tasks, jobs, supervision, hiring, or workflow change around it

- A patent, robot installation, software rollout, and vacancy skill request are different events.

Technology / Innovation → Task content → Skill demand
→ Worker and firm adjustment → Equilibrium outcomes

- Tasks are the bridge from technology to labor demand.
- Skills matter because workers differ in comparative advantage across tasks.
- Firms decide whether to adopt, reorganize, hire, train, substitute, or expand.
- Workers decide whether to learn, move, switch jobs, bargain, or exit.

Substitution, augmentation, and new tasks

Channel	Direct task effect	Typical labor question
Automation	tasks move from labor to capital or software	which workers are displaced?
Augmentation	workers become more productive in existing tasks	whose productivity rises?
New-task creation	labor gains new activities or occupations	where is labor reinstated?

- The same technology can operate through all three channels.

Automation boundary

$$\frac{c_T}{a_{Tk}} \leq \frac{w}{a_{Lk}}$$

- k indexes tasks.
- c_T is the user cost of the technology.
- a_{Tk} and a_{Lk} are technology and labor productivity in task k .
- A shift in this boundary predicts task substitution, not the full equilibrium effect.

Exposure is not effect

$$\Delta Y_{gt} = \beta TechMeasure_{gt} + X'_{gt}\Gamma + \varepsilon_{gt}$$

- What does the technology measure capture: invention, exposure, adoption, diffusion, or realized use?
- What is g : task, occupation, worker, firm, establishment, industry, or place?
- What is Y : wages, employment, vacancies, skill demand, mobility, organization, or welfare?
- Which equilibrium margin is left outside the design?

Measurement is a research frontier

Measure	Closest object	Main warning
Patents and patent text	invention and capability	not realized use
Robots	embodied automation adoption	narrow technology class
Software / IT	digital and organizational adoption	heterogeneous meaning
Vacancy text	desired task and skill demand	postings are not employment
Task surveys / O*NET	job content and exposure structure	can lag current work
Firm investment / resumes	realized adoption and skills	selection and access

Different measures, different objects

- Patent-text exposure can rank occupations by capability similarity before adoption is visible.
- Robot exposure can isolate one concrete embodied automation margin.
- Vacancy text can show changing employer demand for technology-related skills.
- Firm adoption data can move closer to realized use but make selection central.
- Disagreement across measures is often information, not a nuisance.

Main literature map

Literature	Week 1 role
Task content and SBTC	tasks connect technology to labor demand
Automation and new tasks	displacement, productivity, and reinstatement
Robots and ICT	concrete adoption margins
Patent/task exposure	capability-to-task measurement
Vacancies and firm adoption	desired skill demand and realized implementation

- The organizing question is always the same: what is measured and what labor outcome follows?

Research design checklist

- ① Name the technology margin.
- ② State whether the measure captures invention, exposure, adoption, diffusion, or realized use.
- ③ Choose the unit: task, occupation, worker, firm, industry, or place.
- ④ Name the labor outcome and adjustment horizon.
- ⑤ State the equilibrium margin that remains offstage.

- Reproduce: construct a toy technology-to-task exposure score in the spirit of Webb.
- Diagnose: compare patent-text-style exposure, robot-style exposure, and vacancy-style skill demand.
- Transfer: move the logic to synthetic firms with different workforce task portfolios.
- Goal: understand how measurement choices create different empirical conclusions.

- Week 1 builds the common language: technology, tasks, skills, adjustment, equilibrium, and measurement.
- Week 2 studies automation, AI, and labor demand more directly.
- The key next question: which tasks are displaced, complemented, or newly created when technologies diffuse?

Takeaways

- Technology is a labor object when it changes tasks, skills, firms, wages, employment, or welfare.
- Invention, adoption, diffusion, and reorganization are separate empirical margins.
- Automation, augmentation, and new-task creation have different labor-market implications.
- Measurement is not a technical afterthought; it is one of the central open questions in the field.